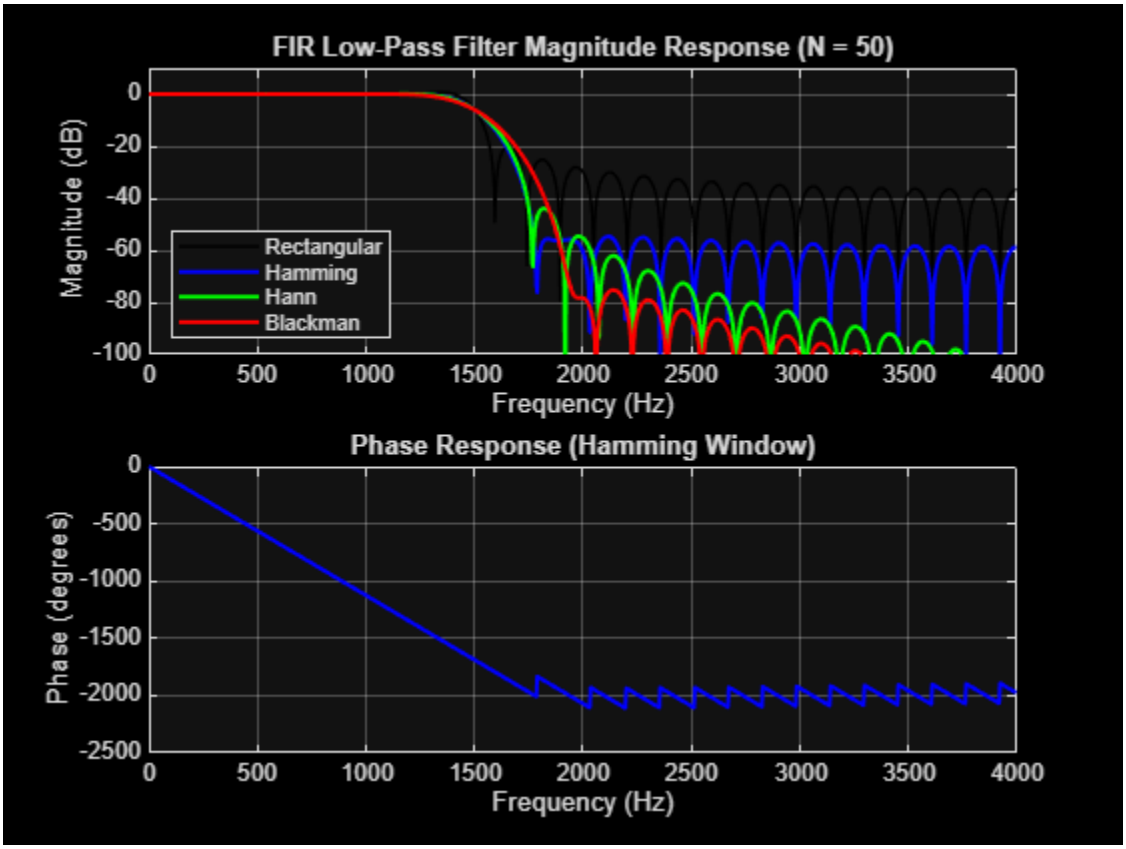
```
clear;
clc;
close all;

Fs = 8000;
fc = 1500;
wc = fc / (Fs/2);
N = 50;

% Window Functions
w_rect  = rectwin(N+1);
w_hamm  = hamming(N+1);
w_hann  = hann(N+1);
w_black = blackman(N+1);

% FIR Low-Pass Filter Design
b_rect  = fir1(N, wc, w_rect);
b_hamm  = fir1(N, wc, w_hamm);
b_hann  = fir1(N, wc, w_hann);
b_black = fir1(N, wc, w_black);

% Frequency Response Calculation
[H_rect, f] = freqz(b_rect, 1, 1024, Fs);
[H_hamm, ~] = freqz(b_hamm, 1, 1024, Fs);
[H_hann, ~] = freqz(b_hann, 1, 1024, Fs);
[H_black, ~] = freqz(b_black, 1, 1024, Fs);

% Display Numerical Outputs
fprintf('=== EXPERIMENT 4: FIR WINDOWING TECHNIQUES ===\n\n');
fprintf('Filter Order (N): %d\n', N);
fprintf('Sampling Frequency: %d Hz\n', Fs);
fprintf('Cutoff Frequency: %d Hz\n\n', fc);

fprintf('--- RECTANGULAR WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_rect(1:5));

fprintf('--- HAMMING WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_hamm(1:5));

fprintf('--- HANNING WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_hann(1:5));

fprintf('--- BLACKMAN WINDOW COEFFICIENTS (First 5) ---\n');
disp(b_black(1:5));

% Convert Magnitude to dB safely
H_rect_dB  = 20*log10(max(abs(H_rect), 1e-10));
H_hamm_dB  = 20*log10(max(abs(H_hamm), 1e-10));
H_hann_dB  = 20*log10(max(abs(H_hann), 1e-10));
H_black_dB = 20*log10(max(abs(H_black), 1e-10));

% Plotting
```

```
figure('Name', 'FIR Windowing Techniques Comparison');

subplot(2,1,1);

plot(f, H_rect_dB, 'k', 'LineWidth', 1);
hold on;
plot(f, H_hamm_dB, 'b', 'LineWidth', 1.5);
plot(f, H_hann_dB, 'g', 'LineWidth', 1.5);
plot(f, H_black_dB, 'r', 'LineWidth', 1.5);

grid on;
legend('Rectangular', 'Hamming', 'Hann', 'Blackman', ...
    'Location', 'southwest');
title('FIR Low-Pass Filter Magnitude Response (N = 50)');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xlim([0 Fs/2]);
ylim([-100 10]);

subplot(2,1,2);

plot(f, unwrap(angle(H_hamm))*180/pi, 'b', 'LineWidth', 1.5);
grid on;
title('Phase Response (Hamming Window)');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);
```

```
=== EXPERIMENT 4: FIR WINDOWING TECHNIQUES ===

Filter Order (N): 50
Sampling Frequency: 8000 Hz
Cutoff Frequency: 1500 Hz

--- RECTANGULAR WINDOW COEFFICIENTS (First 5) ---
   -0.0118         0    0.0128    0.0103   -0.0058

--- HAMMING WINDOW COEFFICIENTS (First 5) ---
   -0.0009         0    0.0012    0.0011   -0.0008

--- HANNING WINDOW COEFFICIENTS (First 5) ---
   1.0e-03 *

         0         0    0.2009    0.3592   -0.3588

--- BLACKMAN WINDOW COEFFICIENTS (First 5) ---
   1.0e-03 *

         0         0    0.0743    0.1374   -0.1434
```



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